Pedro L. Barreto

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ABOUT

Full-stack software developer with hands-on experience building and maintaining applications across cybersecurity, education, and financial administration sectors. Skilled in both front-end and back-end technologies, with a growing focus on computer vision and artificial intelligence through academic and personal projects.

EDUCATION

West Orange High School, Winter Garden, FL, USA (2016 - 2020)

- **GPA:** 3.9 & 4.5 (Weighted)
- **Organizations:** National Honor Society and AFJRTOC FL-802 (4 Years)

University Of Fortaleza, CE, Brazil (2020 - 2025)

- **Bachelor's in Computer Science**

PUC Campinas, SP, Brazil (2026 - 2027)

- **Post-Graduate specialization in Data Science and Machine Learning**

EXPERIENCE

- **Eldorado Research Institute (August 2025 – Present):** Contributing to the development of a corporate platform for an international client, with full-stack responsibilities across backend and frontend layers. Supporting the delivery of scalable, high-impact solutions currently deployed and used globally, with a strong focus on performance, reliability, and user experience.
- **Network Secure (May 2024 – July 2025):** Responsible for the development of Threat Intelligence and Vulnerability modules of the platform used by both analysts and clients, handling front-end and back-end responsibilities to deliver a comprehensive and user-friendly experience. Additionally, I've been tasked to manage a team of three other developers in utilizing the Scrum methodologies to the best of my abilities.
- **University of Fortaleza Tech Dept. (April 2022 – May 2024):** Full-stack software developer at the University of Fortaleza's technology department. I was first part of SISPLAN, the university's budget planning system, which we launched in July 2022. Then, I was instrumental in developing the university's new financial administration system used by employees and students, for which I was responsible for both front-end and back-end features of many parts of the system.
- **Picbuy Internship (June 2021 – January 2022):** This was a startup company that launched an app that provided services to all kinds of stores and restaurants with great client support and analytics. I was able to learn a lot whilst helping them improve their internal administration by creating and implementing functional worksheets and the Pipefy board. Later, I was responsible for client training on the platform and client support.

MAIN SKILLS

Backend:

- Java/Kotlin (Quarkus):
- Databases: *Oracle, Postgres and MongoDB*
- Python (*PyTorch, FastAPI, Streamlit and AirFlow*)
- PL/SQL

Frontend:

- Angular, Thymeleaf, Jquery e Vue.js

Others:

- Docker e Nginx
- Scrum framework
- Kafka & Microservices

Languages:

- English (Native/fluent)
- Portuguese (Native/fluent)
- Spanish (Intermediate)

PERSONAL PROJECTS

- **Convolutional Neural Network-based Real-Time Vehicle Collision Monitoring System (Bachelor's thesis):** Developed, designed, and implemented a real-time surveillance system to detect vehicle collisions using a pre-trained convolutional neural network. The system was trained on a custom dataset with tailored augmentation strategies to improve detection accuracy. Capable of tracking incidents and triggering real-time alerts, the project combines deep learning with computer vision for intelligent traffic monitoring applications.
- ***MaxWatch***: Local-First Smart Home Surveillance Platform (Personal project): Designed and built an edge surveillance system that runs on a home computer, using a refined YOLO neural network to detect people in real time from IP camera streams. Integrated ESP32 sensors (laser tripwire, mmWave presence radar, siren) over an MQTT bus, with fusion rules that trigger alarms and send Telegram alerts with photos. Built the backend in Python (FastAPI, SQLite) and a bilingual Vue 3 web dashboard, packaged with Docker, and added secure remote access over a private VPN so video never leaves the house.

AWARDS

- American Legion JROTC cadet of the year (Post 63) – 2019
- 76th Florida American Legion Boys State – Senator (2019)
- The Air Force Association Award (JROTC National Award) (2020)
- US College Board AP Scholar - 2020